Introduction to Artificial Intelligence

Mihir Rao

Contents

1 Introduction to AI **2**
1.1 Rational decisions 2
1.2 Agents and environments 2
1.3 Methods and applications 3

Lecture 1: Introduction to AI

1.1 Rational decisions

Artificial intelligence can be approached through four goals: making machines think like people, act like people, think rationally, or act rationally. This course focuses on the last: choosing actions that best achieve specified goals. A decision can be rational even if it does not resemble human thinking.

A rational decision maximizes expected utility, a numerical measure of the desirability of an outcome. If h denotes the information available when choosing an action, O is the set of possible outcomes, and $U(o)$ is the utility of outcome o, then for a finite outcome set the rule is

$$a^* \in \arg\max_a \sum_{o \in O} P(o \mid h, a)\, U(o).$$

For each action, weight each possible outcome's utility by its probability, then choose an action with the largest total. Rationality is relative to the goals and information available at decision time: an unlikely bad outcome can occur even when the chosen action had the highest expected utility. The criterion evaluates the decision, not whether its internal reasoning imitates a person.

1.2 Agents and environments

An agent perceives its environment and acts on it. Sensors deliver percepts, and actuators carry out actions. The environment then changes and supplies new percepts, so decisions form a continuing interaction. The nature of the observations, available actions, and environment determines which techniques are appropriate.

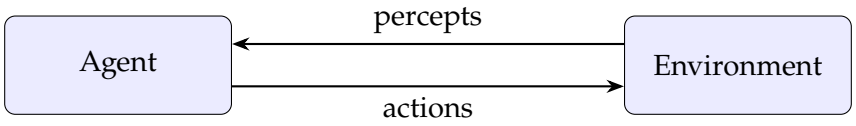


Figure 1: An agent interacts with its environment.

For Pac-Man, the environment is the game, percepts describe what the agent can observe, and actions move Pac-Man through the maze. The agent's objective specifies what counts as a good outcome; seeing the same game does not by itself determine what the agent should prefer.

A REFLEX AGENT chooses from its current percept, possibly using memory, without explicitly evaluating future consequences. A PLANNING AGENT uses a model to consider how candidate actions might change the world and selects a plan in light of its goal. A reflex rule can still be rational when it selects an optimal action for the setting; planning is one way to make decisions, not the definition of rationality.

1.3 Methods and applications

AI systems combine computation and data. Search and planning compare possible actions; constraint satisfaction finds assignments that obey specified rules; adversarial and uncertain settings require reasoning about other agents or uncertain outcomes. Probabilistic models represent uncertainty, decision theory connects uncertainty to utility, and machine learning uses data to improve predictions or decisions.

The same principles appear in different applications:

- **Language:** speech recognition maps audio to text, speech synthesis maps text to audio, and language systems support translation, question answering, and text classification.
- **Vision:** perception turns pixels into information, such as object identities, scene labels, or three-dimensional structure.
- **Robotics:** perception, planning, and control connect decisions to physical actions. The real world can differ from a simulator.
- **Games and logic:** game-playing systems reason over moves; logical systems use deduction, constraint satisfaction, and satisfiability to solve structured problems.

The field has used both knowledge-based and statistical approaches. Recent systems in the 2018 course draw on large datasets, computation, and neural networks; the appropriate representation and objective still depend on the task.

Source. Pieter Abbeel and Dan Klein, Introduction to Artificial Intelligence, Fall 2018, Lecture 1 and Note 1.